Variational and Optimization-based Approaches for Event-Based Vision Applications

From unobserved modes to well-posed estimators

Daniel Blauvelt

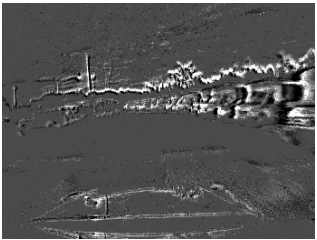
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What can changes tell us about a scene?



A conventional camera records an image.



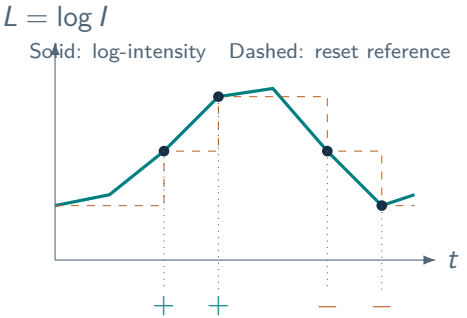
Events can support a reconstructed view.

What determines the answer when the observations leave part of the unknown free?

Takeaway. A mathematically unique estimate can still depend on assumptions the sensor cannot verify.

Dissertation: Chs. 1 and 5; Ch. 2, event-only comparison.

An event camera: each pixel reports threshold crossings



An event is a tuple

$$e_k = (\mathbf{x}_k, s_k, p_k), \qquad p_k = \pm 1.$$

For the ideal threshold model,

$$L(\mathbf{x}_k, s_k) - L(\mathbf{x}_k, s_k^-) = p_k c_{\mathrm{th}}.$$

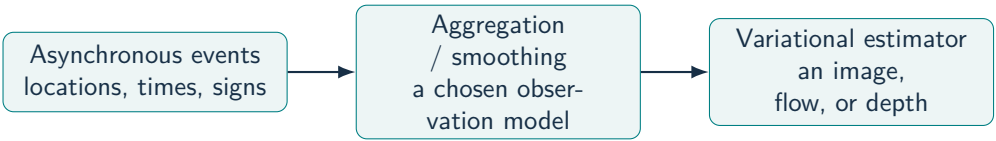
The preceding reset s_k^- is **local to that pixel**.

Takeaway. Events record changes. Adding a constant to a pixel's log-intensity leaves its events unchanged.

Dissertation: Ch. 2, eq:polarity; Ch. 3, eq:event_integral. Diagram is schematic.

A point measure needs a representation before optimization

$$\mathcal{E} = \sum_k p_k \, \delta_{(\mathbf{x}_k, s_k)}.$$



- Temporal kernels yield integrated changes at reconstruction times.
- Time surfaces and voxel grids yield coefficients for motion constraints.
- Stereo accumulation yields paired temporal event-feature observations.

Takeaway. The representation supplies usable coefficients and introduces modeling choices.

Dissertation: Ch. 1; Ch. 2, eq:exy–eq:Exy; Ch. 3, sec:framework; Ch. 4, sec:normalization.

The dissertation's progression

Problem	What the data leave free	How the model controls it
Temporal reconstruction	Additive log-intensity baseline	Positive magnitude penalty; unique fixed-weight solve
Dense optical flow	Motion tangent to an edge	Spatial smoothness plus integrated directional information
Stereo depth	Competing correspondence candidates	Coupled profile solves; separate cost-gap conditions

Three recurring questions: **existence**, **stability**, and **physical identifiability**.

Dissertation: Ch. 5, tab:contribution_matrix.

Start with one pixel: integrated changes on a temporal chain

Choose a unit-mass temporal kernel ϕ_{s_k}:

$$e(t) = \sum_{k \text{ at this pixel}} p_k \phi_{s_k}(t).$$

$$y_i = \int_{t_i}^{t_{i+1}} e(t)\, dt.$$

The reconstruction times t_i are chosen by the algorithm.

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix}.$$

$$\mathbf{A}\mathbf{f} \approx \mathbf{y}.$$

$$\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}.$$

Events determine increments, leaving a constant free.

Takeaway. Kernel width controls temporal averaging; a baseline requires an additional choice.

Dissertation: Ch. 2, eq:exy, eq:Exy, eq:tempdiffA, lem:nullspace.

A regularized estimate is uniquely selected

For fixed $\mathbf{\Lambda} = \mathrm{diag}(\lambda_i)$ with every $\lambda_i > 0$,

$$\mathbf{f}^* = \arg\min_{\mathbf{f}\in\mathbb{R}^n} \left\{ \frac{1}{2} \|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \frac{1}{2} \|\mathbf{\Lambda}\mathbf{f}\|^2 \right\}.$$

Temporal uniqueness theorem

$$\mathbf{K}_{\mathrm{time}} = \mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda} \succ 0, \qquad \mathbf{K}_{\mathrm{time}} \mathbf{f}^* = \mathbf{A}^\top \mathbf{y}.$$

The fixed-weight problem has a unique global minimizer.

The selected baseline satisfies $\mathbf{1}^\top \mathbf{\Lambda}^2 \mathbf{f}^* = 0$.

Takeaway. The penalty chooses an unobserved baseline and also changes observable increments.

Dissertation: Ch. 2, eq:in, thm:uniqueness. Baseline identity: direct consequence.

Why uniqueness and stability hold

Write $\lambda_{\min} = \min_i \lambda_i$. For $\mathbf{z} \neq 0$,

$$\mathbf{z}^\top \mathbf{K}_{\text{time}} \mathbf{z} = \|\mathbf{A}\mathbf{z}\|^2 + \|\mathbf{\Lambda}\mathbf{z}\|^2 \geq \lambda_{\min}^2 \|\mathbf{z}\|^2 > 0.$$

At the same fixed weights,

$$\delta\mathbf{f}^* = \mathbf{K}_{\text{time}}^{-1} \mathbf{A}^\top \delta\mathbf{y}, \qquad \|\delta\mathbf{f}^*\| \leq \frac{\|\mathbf{A}\|_2}{\lambda_{\min}^2} \|\delta\mathbf{y}\|.$$

$$\|\mathbf{A}\|_2 \leq 2, \qquad \kappa_2(\mathbf{K}_{\text{time}}) \leq \frac{4 + \lambda_{\max}^2}{\lambda_{\min}^2}.$$

Takeaway. Coercivity gives a stable estimate; weakening the penalty weakens these full-space bounds.

Dissertation: Ch. 2, thm:uniqueness, prop:stability.

How are the temporal weights chosen?

For activity $a_i = |dc_i|$, the selected sigmoid rule is

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}}.$$

Here dc_i is the signed event accumulation; a_i is its magnitude, not a count of all events.

Rule	Sigmoid	Inverse proportional	Exponential
Outdoor condition number	29.60	81.50	100.74

Chosen empirically to balance image fidelity and conditioning.

Takeaway. Positive weights ensure a unique solve; the stability bound holds with weights fixed.

Dissertation: Ch. 2, subsubsec:lambdacal, subsubsec:lambda_comparison, tab:lambda_outdoor_avg.

Relation to Antil–Sayre bilevel reconstruction

This chapter
Events only $\longrightarrow$ activity-based $\Lambda(dc)$ $\longrightarrow$ fixed convex temporal solve.

Antil–Sayre (2023)
Events and intensity frames $\longrightarrow$ outer optimization over threshold parameters z.

Sequence	Method	PSNR	CN
Indoor	Bilevel / event-only	15.37 / 10.85	39755 / 25.51
Outdoor	Bilevel / event-only	8.41 / 8.71	39755 / 27.10

PSNR: reference-image similarity (dB). CN: reported system condition number.

Takeaway. Frame information and a different optimization problem change the comparison.

Dissertation: Antil–Sayre (2023), Sec. 3, Eqs. (3.1)–(3.3); Ch. 2, subsubsec:bilevel_comparison, tab:bilevel_comp_indoor, tab:bilevel_comp_outdoor.

Shared motion couples temporal profiles to one scene

For n frames, stack the fixed, row-normalized known-motion footprint projections P_k into P; $P_k\mathbf{1} = \mathbf{1}$. With $A \in \mathbb{R}^{(n-1)\times n}$ the first-difference matrix,

$$B_{\mathrm{SAR}} = (A \otimes I)P, \qquad \ker B_{\mathrm{SAR}} = \{\theta : P_{k+1}\theta = P_k\theta,\ 1 \le k < n\}, \qquad B_{\mathrm{SAR}}\mathbf{1} = 0.$$

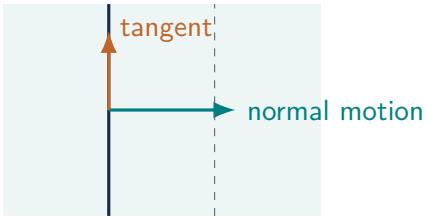
With an unsquared coefficient $\lambda > 0$,

$$\left(B_{\mathrm{SAR}}^{\top} B_{\mathrm{SAR}} + \lambda I\right) \widehat{\theta} = B_{\mathrm{SAR}}^{\top} b.$$

Takeaway. Motion links per-frame changes to a shared scene while retaining motion-invariant scene modes.

Dissertation: Ch. 2, prop:ch2_sar_solvability.

Motion introduces a new invisible direction



A straight edge reveals its normal speed.

A time surface $\mathcal{T}(\mathbf{x})$ records an edge's crossing time.

$$\nabla \mathcal{T} \cdot \mathbf{v} = 1.$$

One scalar equation constrains a vector velocity.

$$\mathbf{v} = \frac{\nabla \mathcal{T}}{|\nabla \mathcal{T}|^2} + \tau \nabla \mathcal{T}^{\perp}.$$

Takeaway. This is the aperture problem: an edge does not reveal motion along its tangent.

Dissertation: Ch. 3, prop:time_surface and following aperture remark.

Derive the time-surface constraint by following the edge

For an ideal single crossing, define

$$\psi(\mathbf{x}, t) = t - \mathcal{T}(\mathbf{x}), \qquad \psi(\mathbf{x}(t), t) = 0.$$

If the edge translates with constant $\dot{\mathbf{x}} = \mathbf{v}$, then

$$0 = \frac{d}{dt}\psi(\mathbf{x}(t), t) = 1 - \nabla\mathcal{T} \cdot \mathbf{v}.$$

Model hypotheses

The edge crosses each location once, has nonzero normal speed, and its recorded timestamp equals a differentiable crossing-time map.

Takeaway. Interpolation, stale timestamps, and repeated crossings make this an approximate data constraint.

Dissertation: Ch. 3, prop:time_surface; def:time_surface.

Time-surface gradient estimation: bias versus variance

Fit a plane to timestamps at offsets $\mathbf{z}_j$ around a pixel.

$$Q = \sum_j \mathbf{z}_j \mathbf{z}_j^\top \succ 0.$$

For a centrally symmetric design and independent timestamp errors with common mean and finite variance σ^2:

$$\mathrm{Cov}(\widehat{\nabla \mathcal{T}}) = \sigma^2 Q^{-1}.$$

With a Lipschitz Hessian of constant L_H on a neighborhood containing the sample disk,

$$\|\mathrm{Bias}\| \le \frac{L_H}{6} \left\|Q^{-1}\right\| \sum_j \|\mathbf{z}_j\|^4, \qquad \mathrm{MSE} = \|\mathrm{Bias}\|^2 + \sigma^2 \,\mathrm{tr}(Q^{-1}).$$

Takeaway. A larger neighborhood reduces variance but can increase curvature bias. Regression rank and velocity identifiability are different questions.

Dissertation: Ch. 3, prop:ts_regression_bound.

Voxel projection: transport a represented event-activity field

Smooth and bin events into a spatiotemporal field $V(\mathbf{x}, t)$.

$$a(\mathbf{x}, s) = \int_{s}^{s+T} V(\mathbf{x}, t)\, dt.$$

If $V_t + \mathbf{v} \cdot \nabla V = 0$ with constant velocity over the window,

$$a_s + \mathbf{v} \cdot \nabla a = 0, \qquad a_s = V(\mathbf{x}, s+T) - V(\mathbf{x}, s).$$

The voxel-projection residual is $\nabla a \cdot \mathbf{v} + a_s$.

Takeaway. Activity is a count-derived field. Treating it as transported is an additional modeling assumption.

Dissertation: Ch. 3, subsec:vox_projection; derivative-operator qualifications.

The structure tensor pools directional information

At a fixed evaluation time $t_\star$, set

$$\mathbf{J}(\mathbf{x}) = \int W(\mathbf{x}-\mathbf{y})\, \nabla_3 V(\mathbf{y}, t_\star) \nabla_3 V(\mathbf{y}, t_\star)^\top \, d\mathbf{y}, \qquad W \geq 0.$$

Use the local residual energy

$$\mathbf{v}_3^\top \mathbf{J} \mathbf{v}_3, \qquad \mathbf{v}_3 = (v_x, v_y, 1)^\top, \qquad \mathbf{J} = \begin{pmatrix} \widetilde{\mathbf{J}} & \mathbf{j}_t \\ \mathbf{j}_t^\top & J_{tt} \end{pmatrix}.$$

- Each gradient outer product is positive semidefinite and rank at most one.
- Pooling nonparallel spatial gradients can make $\widetilde{\mathbf{J}}$ full rank.

Takeaway. Pooling can identify both components locally; the global theorem needs only integrated identifiability.

Dissertation: Ch. 3, subsec:vox_struct, prop:vox_tensor_wellposed.

Three representations, one quadratic variational problem

Rename the flow data matrix $\mathbf{B}$ to distinguish it from temporal $\mathbf{A}$:

$$E_{\mathbf{B},b}(\mathbf{v}) = \frac{1}{2}\int_\Omega (\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}) + \frac{\alpha^2}{2}\int_\Omega |\nabla \mathbf{v}|^2.$$

Representation	$\mathbf{B}$	b
Time surface	$\nabla\mathcal{T}\nabla\mathcal{T}^\top$	$\nabla\mathcal{T}$
Voxel projection	$\nabla a \nabla a^\top$	$-a_s\nabla a$
Structure tensor	$\widetilde{\mathbf{J}}$	$-\mathbf{j}_t$

Takeaway. The representations change the coefficients; the analysis can be proved once.

Dissertation: Ch. 3, thm:unified_wellposedness and its three corollaries.

The main flow theorem: global information controls constants

Unified well-posedness theorem

Let $\Omega \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz; $\alpha > 0$. Assume

$$\mathbf{B} \in L^\infty, \quad \mathbf{B} = \mathbf{B}^\top \succeq 0 \text{ a.e.}, \quad b \in L^2, \qquad \int_\Omega \mathbf{B}(\mathbf{x})\, d\mathbf{x} \succ 0.$$

Then $E_{\mathbf{B},b}$ has a unique minimizer in $H^1(\Omega;\mathbb{R}^2)$.

The integrated matrix must identify **every constant velocity direction**.

Takeaway. B may vanish on open regions; integrated directional information and smoothness still give a unique estimate.

Dissertation: Ch. 3, thm:unified_wellposedness, eq:integrated_identifiability.

Proof: control the mean and the fluctuation separately

Decompose $\mathbf{v} = \bar{\mathbf{v}} + \mathbf{w}$, with $\int_\Omega \mathbf{w} = 0$.

$$\|\mathbf{w}\|_{L^2} \le C_P \, \|\nabla \mathbf{v}\|_{L^2} .$$

Let $\beta_B = \|\mathbf{B}\|_{L^\infty}$ and $m = \lambda_{\min}(\int_\Omega \mathbf{B}) > 0$.

$$m|\bar{\mathbf{v}}|^2 \le 2 \int_\Omega \mathbf{v}^\top \mathbf{B} \mathbf{v} + 2\beta_B C_P^2 \, \|\nabla \mathbf{v}\|_{L^2}^2 .$$

Therefore the bilinear form satisfies

$$\mathcal{B}(\mathbf{v}, \mathbf{v}) = \int \mathbf{v}^\top \mathbf{B} \mathbf{v} + \alpha^2 \int |\nabla \mathbf{v}|^2 \ge c_0 \, \|\mathbf{v}\|_{H^1}^2 , \qquad c_0 = c_0(\Omega, \alpha, \beta_B, m).$$

Takeaway. Poincaré controls fluctuations, data control the mean, and Lax–Milgram supplies the unique weak solution.

Dissertation: Ch. 3, proof of thm:unified_wellposedness.

Stability follows by subtracting weak equations

Suppose two coefficient pairs share bounds

$$\|\mathbf{B}_i\|_{L^\infty} \le \beta_B, \qquad \lambda_{\min}\left(\int \mathbf{B}_i\right) \ge m > 0, \qquad \|b_i\|_{L^2} \le R_b.$$

Here $c_0 = c_0(\Omega, \alpha, \beta_B, m)$ is independent of b_i. For $\mathbf{z} = \mathbf{v}_1^* - \mathbf{v}_2^*$,

$$\mathcal{B}_1(\mathbf{z}, \mathbf{z}) = \int (b_1 - b_2) \cdot \mathbf{z} + \int \mathbf{z}^\top (\mathbf{B}_2 - \mathbf{B}_1)\mathbf{v}_2^*.$$

Hence

$$\|\mathbf{z}\|_{H^1} \le c_0^{-1} \|b_1 - b_2\|_{L^2} + c_0^{-2} R_b \|\mathbf{B}_1 - \mathbf{B}_2\|_{L^\infty}.$$

Takeaway. The estimator is Lipschitz in its coefficients while identifiability remains uniformly positive.

Dissertation: Ch. 3, thm:unified_wellposedness, eq:abstract_stability.

Discrete flow: connectivity and sampled directions matter

For $\alpha > 0$, symmetric nonnegative graph weights with degrees $d_i > 0$, and $\mathbf{B}_i = \mathbf{B}_i^\top \succeq 0$, the quadratic curvature is

$$\mathbf{H}_h = \operatorname{diag}(q_i \mathbf{B}_i) + \alpha^2 (L_G \otimes \mathbf{I}_2), \qquad q_i > 0.$$

Discrete uniqueness and block-Jacobi convergence

If the graph is connected and $\sum_i q_i \mathbf{B}_i \succ 0$, then the discrete energy has a unique minimizer. The corresponding simultaneous block-Jacobi sweep converges from every initial field.

- Zero graph energy forces a constant field; the sampled Gram matrix eliminates it.
- For Jacobi, both $D - R \succ 0$ and $D + R \succ 0$ put its spectrum inside $(-1, 1)$.

Dissertation: Ch. 3, cor:vox_projection_wellposed; positive-q_i extension of thm:convergence and cor:convergence_psd_blocks.

Contrast maximization changes the optimization problem

Warp each event to a common normalized reference time:

$$\mathbf{x}_k^w = \mathbf{x}_k + (\widetilde{t}_{\mathrm{ref}} - \widetilde{t}_k)\mathbf{q}(\mathbf{x}_k), \qquad \mathbf{q} = \mathcal{T}\mathbf{v}_{\mathrm{phys}}.$$

Represent $\mathbf{q}$ by continuous B-splines with coefficients $\mathbf{c}$.

$$F(\mathbf{c}) = -\operatorname{Var}(I_{\mathbf{c}}) + \mathbf{c}^\top \mathbf{K}_{\mathrm{reg}}\mathbf{c}, \qquad \mathbf{K}_{\mathrm{reg}} \succeq \tau_{\mathrm{ridge}}\mathbf{I}, \quad \tau_{\mathrm{ridge}} > 0.$$

- A good warp concentrates aligned events.
- A flexible incorrect warp can also concentrate events by collapsing them.

Takeaway. The full objective is generally nonconvex. The quadratic flow theorem does not apply.

Dissertation: Ch. 3, sec:cmax, prop:cmax_wellposed, prop:cmax_existence.

A global minimizer exists, even though the solver is local

For finitely many fixed event weights, let $W_{\mathrm{tot}} = \sum_k |w_k|$. The bilinear accumulation is continuous and satisfies

$$\mathrm{Var}(I_{\mathbf{c}}) \le \frac{W_{\mathrm{tot}}^2}{HW}.$$

Therefore

$$F(\mathbf{c}) \ge -\frac{W_{\mathrm{tot}}^2}{HW} + \tau_{\mathrm{ridge}} \|\mathbf{c}\|^2 \longrightarrow \infty.$$

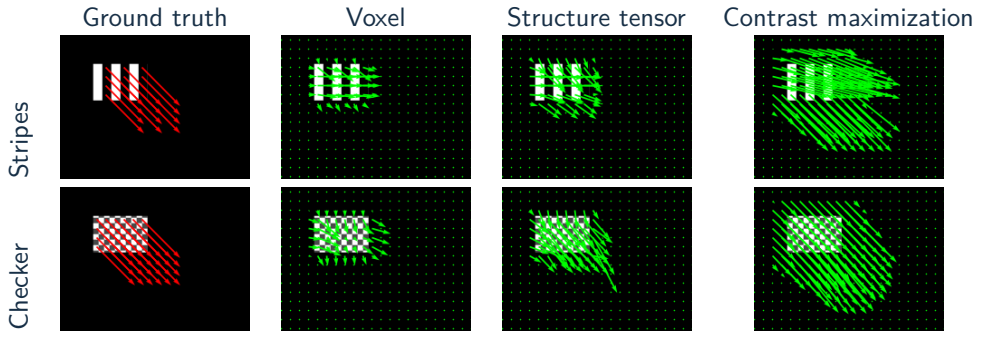
Proof mechanism

Continuous objective + bounded focus gain + coercive ridge $\Rightarrow$ compact sublevel set $\Rightarrow$ Weierstrass.

Takeaway. A global minimizer exists by finite-dimensional coercivity; the local solver has no global guarantee.

Dissertation: Ch. 3, prop:cmax_existence and evaluated-solver scope.

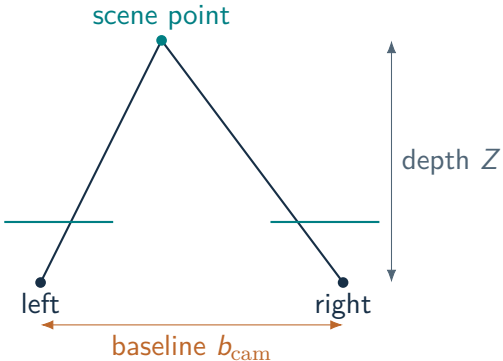
Known-motion examples make the aperture problem visible



Takeaway. Stripes emphasize a normal-flow constraint; checker texture supplies directions in both axes.

Dissertation: Ch. 3, fig:ch3_quiver_v2e; v2e simulation, same window and arrow scale.

Two viewpoints turn correspondence into depth



Rectification makes the correspondence search horizontal:

$$d = x_L - x_R.$$

$$Z = \frac{f_{\mathrm{cam}} b_{\mathrm{cam}}}{d}, \qquad d > 0.$$

Each candidate d pairs a left observation with $\mathbf{y}_R(x - d, y)$.

Takeaway. The correspondence is unknown; solving the latent profiles at a fixed candidate is only the inner problem.

Dissertation: Ch. 4, sec:solution, eq:depth_conversion. Geometry is schematic.

At each candidate, couple two temporal profiles

Suppress the pixel index; $\mathbf{y}_R = \mathbf{y}_R(d)$.

$$J_d(\mathbf{f}_L, \mathbf{f}_R; \mathbf{W}_L, \mathbf{W}_R) = \frac{1}{2}\left\|\mathbf{W}_L^{1/2}(\mathbf{A}\mathbf{f}_L - \mathbf{y}_L)\right\|^2 + \frac{1}{2}\left\|\mathbf{W}_R^{1/2}(\mathbf{A}\mathbf{f}_R - \mathbf{y}_R)\right\|^2 + \frac{1}{2}\left\|\mathbf{\Lambda}\mathbf{f}_L\right\|^2 + \frac{1}{2}\left\|\mathbf{\Lambda}\mathbf{f}_R\right\|^2 + \frac{\gamma}{2}\left\|\mathbf{f}_L - \mathbf{f}_R\right\|^2.$$

- Data fidelity estimates each temporal profile from its own view.
- Magnitude penalties control temporal baselines.
- Soft coupling penalizes left–right disagreement.

Takeaway. The right latent profile is auxiliary to this pixel and candidate, rather than a single globally shared right image.

Dissertation: Ch. 4, eq:weighted_objective; fixed-disparity model definition.

Frozen weights give a unique, stable inner solve

Fix d, $\mathbf{W}_L, \mathbf{W}_R \succ 0$, $\mathbf{\Lambda} \succ 0$, and $\gamma \geq 0$. Define

$$\mathbf{K}_c = \mathbf{A}^\top \mathbf{W}_c \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}, \qquad \mathbf{M}_W = \begin{pmatrix} \mathbf{K}_L + \gamma \mathbf{I} & -\gamma \mathbf{I} \\ -\gamma \mathbf{I} & \mathbf{K}_R + \gamma \mathbf{I} \end{pmatrix}.$$

For $\mathbf{z} = (\mathbf{z}_L, \mathbf{z}_R)$,

$$\mathbf{z}^\top \mathbf{M}_W \mathbf{z} = \sum_{c=L,R} \left(\left\| \mathbf{W}_c^{1/2} \mathbf{A} \mathbf{z}_c \right\|^2 + \left\| \mathbf{\Lambda} \mathbf{z}_c \right\|^2 \right) + \gamma \left\| \mathbf{z}_L - \mathbf{z}_R \right\|^2.$$

Conclusion

$\mathbf{M}_W \succeq \lambda_{\min}^2 \mathbf{I} \succ 0$: the frozen quadratic has exactly one global minimizer.

Dissertation: Ch. 4, eq:mw_quadform, thm:weighted_spd, thm:convexity.

Sum and difference modes expose the coupling

For equal blocks $\mathbf{K}_L = \mathbf{K}_R = \mathbf{K}$, introduce

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{I} & -\mathbf{I} \end{pmatrix}, \qquad \mathbf{U}^\top \mathbf{M} \mathbf{U} = \begin{pmatrix} \mathbf{K} & 0 \\ 0 & \mathbf{K} + 2\gamma \mathbf{I} \end{pmatrix}.$$

Mean profile
The coupling cancels. Shared information is fused by the temporal solve.

Difference profile
The coupling raises curvature and suppresses disagreement.

For $\mathbf{\Lambda} = \lambda \mathbf{I}$ in the unweighted model,

$$\kappa_2(\mathbf{M}) = 1 + \frac{g_{\max} + 2\gamma}{\lambda^2}.$$

Dissertation: Ch. 4, lem:block_diag_M, thm:conditioning.

Profile out the latent variables to see what disparity compares

In the unweighted model, let $\mathbf{K} = \mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}$ and

$$m(\mathbf{y}) = \frac{1}{2} \mathbf{y}^\top (\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top)\mathbf{y}.$$

The exact minimized candidate cost is

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_R(d)) + \frac{1}{4}\boldsymbol{\delta}_d^\top \left[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}\right] \boldsymbol{\delta}_d, \quad \boldsymbol{\delta}_d = \mathbf{A}^\top(\mathbf{y}_L - \mathbf{y}_R(d)).$$

- The final term measures left–right discrepancy.
- The right monocular cost also changes with the candidate.

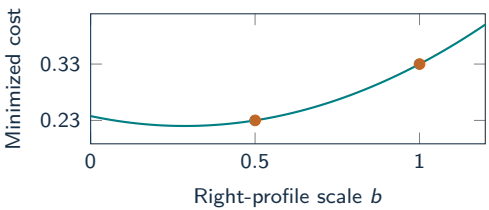
Takeaway. The minimized regularized energy is more than a correspondence discrepancy.

Dissertation: Ch. 4, lem:matching_cost_closed_form, prop:global_opt_unweighted. Equal fixed weights admit a scaled analogue.

A perfect correspondence need not have the lowest cost

Take $\mathbf{A} = [-1, 1]$, $\mathbf{\Lambda} = \mathbf{I}$, $\gamma = 1$, $y_L = 1$, and $y_R = b$.

$$C(b) = \frac{1 + b^2}{6} + \frac{(1 - b)^2}{15}.$$



Let D denote the discrepancy term in the preceding profiled-cost formula.

$$C(1) = \frac{1}{3}, \qquad C\left(\frac{1}{2}\right) = \frac{9}{40} < \frac{1}{3}.$$

$$b^* = \frac{D(y_L)}{m(y_L) + D(y_L)} = \frac{2}{7}.$$

An attenuated wrong candidate beats the exact match if both are available.

Takeaway. Inner uniqueness, distinct observations, and correct correspondence are separate claims.

Dissertation: Ch. 4, prop:attenuation_bias, sec:identifiability. Unweighted cost before aggregation.

Correct disparity needs a separation condition

Write the unweighted cost as

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_d) + D(\mathbf{y}_L - \mathbf{y}_d), \qquad D(\mathbf{e}) \geq 0.$$

A sufficient correct-selection theorem

On a finite candidate set, suppose $\mathbf{y}_{d_*} = \mathbf{y}_L$, the positive regularizer is common to both views and all candidates, and $\gamma > 0$ is candidate-independent. If every competitor has

$$\mathbf{y}_d \neq \mathbf{y}_L \quad \text{and} \quad m(\mathbf{y}_d) \geq m(\mathbf{y}_L),$$

then $C(d) > C(d_*)$: the correct candidate is the unique winner.

Takeaway. The discrepancy must overcome candidate-dependent monocular bias. Stronger metric-separation criteria are also available.

Dissertation: Ch. 4, prop:correct_selection(b1); additional criterion in backup.

A winning margin turns cost stability into selection stability

For a fixed finite set of at least two candidates with a unique winner d_*, define

$$\Delta = \min_{d \neq d_*} \left(C(d) - C(d_*) \right) > 0.$$

If $|\widetilde{C}(d) - C(d)| \leq \eta$ for all candidates, including d_*, then every $d \neq d_*$ satisfies

$$\widetilde{C}(d) - \widetilde{C}(d_*) \geq \Delta - 2\eta > 0 \qquad \text{when } \eta < \Delta/2.$$

A non-identifiability example

Two candidates with identical right-view profiles have identical local costs when the candidate procedures and parameters coincide.

Takeaway. Temporal activity alone cannot guarantee correspondence separation.

Dissertation: Ch. 4, cor:selection_robust, prop:disparity_nonidentifiability.

Heteroscedastic weights: statistical motivation and scope

As a working raw-count model,

$$N_+ \sim \text{Poisson}(\theta_+), \qquad N_- \sim \text{Poisson}(\theta_-),$$

independently. For $b = N_+ - N_-$,

$$\mathbb{E}b = \theta_+ - \theta_-, \qquad \text{Var}(b) = \theta_+ + \theta_-.$$

$$|\theta_+ - \theta_-| \leq \theta_+ + \theta_-, \quad \text{with equality iff at least one rate is zero.}$$

The evaluated reciprocal proxy is

$$W_{ii}^{(k)} = \frac{1}{|(\mathbf{A}\mathbf{f}^{(k)})_i| + \varepsilon_{\text{noise}}}, \qquad \varepsilon_{\text{noise}} > 0.$$

Takeaway. Net rate does not determine total variance. The processed-data weights define a frozen Gaussian surrogate.

Dissertation: Ch. 4, def:skellam_surrogate and following scope discussion.

IRLS: fixed-point existence and conditional convergence

Fix d, $\mathbf{\Lambda}$, and γ; stack $\mathbf{y} = (\mathbf{y}_L, \mathbf{y}_R)$. Let $\Phi(\mathbf{g})$ minimize the quadratic with weights frozen at $\mathbf{g}$. For the *unclipped reciprocal map*, set $\ell = \lambda_{\min}^2 > 0$:

$$\|\Phi(\mathbf{g})\|^2 \le \frac{\|\mathbf{y}\|^2}{\ell\, \varepsilon_{\text{noise}}}.$$

A sufficiently large closed ball is invariant; continuity and Brouwer give a fixed point.

Sufficient convergence condition

$$q = \frac{\|\mathbf{A}\|_2\, \|\mathbf{y}\|}{\sqrt{\ell}\, \varepsilon_{\text{noise}}^{3/2}} < 1 \quad \Rightarrow \quad \text{unique fixed point and geometric convergence.}$$

For fixed damping $\omega \in (0, 1]$, the bound is $q_\omega = 1 - \omega + \omega q$.

Takeaway. A fixed point exists; convergence follows under the displayed sufficient condition.

Dissertation: Ch. 4, lem:fixed_point_existence, prop:irls_contraction, sec:irls_scope.

Stereo benchmark: read accuracy alongside scored coverage

MVSEC indoor flying; raw outputs, including finest-level score interpolation.

Method	Mean depth error (cm) ↓				log RMSE ×100 ↓	Scored points (M)
	FI1	FI2	FI3	FI4		
Unweighted control	140.23	115.40	142.35	113.00	96.18	1.43
Unweighted SGM	83.45	72.90	85.33	71.62	50.55	2.39
Soft-coupled WLS	73.83	**58.35**	70.33	59.24	**44.54**	2.66
Piatkowska adaptation	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20

Each method is scored on its own finite estimates and ground truth in $[0.5, 8]$ m. Point counts are summed across sequences; log RMSE is averaged across sequences.

Takeaway. WLS is competitive with a larger scored population. This table does not compare accuracy on a common pixel set.

Dissertation: Appendix, tab:unified_benchmark, tab:metrics; external rows are local adaptations.

A matched ablation supports the selected weighting rule

Compare reciprocal versus rate-proportional weights while holding the remaining configuration fixed.

	Flying 1	Flying 2	Flying 3	Flying 4
Mean-error reduction (cm)	4.96	8.54	2.90	5.56

The reciprocal rule improved every reported metric on every sequence in this common-set comparison.

Scope of this evidence

Both methods use their common valid pixel set. This is a reanalysis of the same benchmark run, and the rule was selected on these sequences.

Takeaway. A controlled weighting comparison supports the chosen rule; it does not validate the variance proxy or isolate all configuration improvements.

Dissertation: Appendix, sec:success_wls.

Hard fusion changes both the constraint and the regularization

Joint intensity fusion (JIF) imposes $\mathbf{f}_L = \mathbf{f}_R = \mathbf{v}$ and solves

$$(2\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)\mathbf{v} = \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

For fixed equal weights, the unchanged soft model has the limit

$$\gamma \to \infty : \qquad \mathbf{f}_L, \mathbf{f}_R \to (\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)^{-1} \frac{1}{2} \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

JIF instead uses $\mathbf{A}^\top \mathbf{W}\mathbf{A} + \frac{1}{2}\Lambda^2$ after dividing its equation by two.

Takeaway. Reported mean depth errors: JIF 102.10 cm; soft WLS 65.44 cm. Hard equality, regularization, weights, iteration budgets, and scored populations differ.

Dissertation: Ch. 4, prop:jif_relation, sec:jif_negative_result.

What is proved, and what remains a modeling question

Problem	Established mathematical result	Additional physical / algorithmic question
Temporal	Unique fixed-weight minimizer and perturbation bounds	Absolute baseline; adaptive-weight stability
Quadratic flow	H^1 well-posedness; coefficient stability; scoped graph convergence	Representation accuracy and motion boundaries
Contrast max.	Global minimizer exists with coefficient ridge	Nonconvex optimizer convergence and selection
Stereo	Frozen SPD solves; fixed-point existence; conditional contraction; ideal selection margins	Full IRLS stopping rules; weighted, aggregated correspondence

Takeaway. Regularization makes an estimator determinate under stated hypotheses. Physical identifiability remains a separate requirement.

Dissertation: Ch. 5, tab:contribution_matrix.

Next questions follow directly from the proved bounds

1. **Quantify the represented observation error.**
   Separate timestamp estimation, binning, smoothing, and sensor uncertainty.
2. **Close the adaptive-estimator gap.**
   Analyze weight recomputation, stopping rules, and cost perturbations together.
3. **Extend correspondence separation.**
   Move from ideal unweighted costs to weighted and aggregated costs.
4. **Test modeling choices under matched conditions.**
   Use common pixel populations, controlled fusion, and event-active residual diagnostics.

Dissertation: Ch. 5, Future Directions; Ch. 3 residual diagnostic; Ch. 4 selection and IRLS scope.

From event changes to well-posed estimators

Temporal: control an unobserved baseline.

Spatial: combine directional constraints globally.

Binocular: separate inner uniqueness from correspondence.

The common mathematical question is:

Which directions do the observations control, and which are selected by the model?

Thank you. Discussion and additional proofs follow.

Discussion guide

Observation and temporal inversion

- Minimal uniqueness condition / gauge
- Measure-valued continuous-time limit
- Signal gain versus noise gain
- Kernel comparison on indoor and outdoor data

Flow analysis

- Weak Euler equation and boundary condition
- Exact block-Jacobi spectral argument
- Anisotropy and ellipticity
- Regression design and projection mismatch

Stereo analysis

- Mean-zero conditioning
- Cost ellipsoid and observation perturbations
- IRLS contraction proof
- Fixed points versus likelihood stationarity
- Implemented scoring and aggregation

Evidence and interpretation

- Depth evaluation, coverage, and selection
- What is contributed by the dissertation?
- Literature anchors and source map

Must every temporal penalty entry be positive?

For arbitrary $\mathbf{\Lambda}$, the exact algebraic criterion is

$$\mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda} \succ 0 \quad \Longleftrightarrow \quad \ker \mathbf{A} \cap \ker \mathbf{\Lambda} = \{0\}.$$

Why?

The quadratic form vanishes precisely when both $\mathbf{A}\mathbf{z} = 0$ and $\mathbf{\Lambda}\mathbf{z} = 0$.

On a connected first-difference chain, $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$.

- A single nonzero diagonal penalty entry removes that constant mode.
- Fixing one reference value or imposing zero mean also removes it.

Takeaway. A gauge resolves the mathematical ambiguity; a measured reference would add physical information.

Dissertation: Ch. 2, lem:nullspace; Ch. 4, discussion following lem:A_spectrum.

The event measure has a jump primitive

At one pixel, set $\mu = \sum_j p_j \delta_{s_j}$, and choose a consistent right-continuous primitive

$$g(t) = f_0 + \mu((t_0, t]), \qquad g(b) - g(a) = \mu((a, b]).$$

For a standard smooth compactly supported unit-mass kernel,

$$e_h = \mu * \phi_h, \qquad g_h(t) = f_0 + \int_{t_0}^{t} e_h(\xi)\, d\xi.$$

For finitely many interior events, $g_h \to g$ in L^1 as $h \downarrow 0$.

- g belongs to BV; a nonzero interior jump prevents $g \in H^1$.
- The displayed g is an unregularized cumulative-event surrogate.

Takeaway. Keep interval conventions consistent and use increments, rather than an unscaled derivative, in $\mathbf{Af} \approx \mathbf{y}$.

Dissertation: Ch. 2, subsubsec:ode; endpoint convention made explicit here.

Latent-signal gain and observation-noise gain are different

For the unweighted monocular model with scalar $\lambda > 0$,

$$\mathbf{f}^* = (\mathbf{A}^\top \mathbf{A} + \lambda^2 \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{y}.$$

Input	Output modal gain	Interpretation
Latent signal	$g_k/(g_k + \lambda^2)$	Low modes attenuated most
Difference observation	$\sqrt{g_k}/(g_k + \lambda^2)$	Noise amplitude transfer
White difference noise	$\sigma^2 g_k/(g_k + \lambda^2)^2$	Reconstructed modal variance

$$\max_{g \geq 0} \frac{g}{(g + \lambda^2)^2} = \frac{1}{4\lambda^2}.$$

Dissertation: Ch. 4, thm:frequency_response, prop:noise_gain.

Kernel performance depends on the scene and adaptive weights

	MVSEC indoor		MVSEC outdoor	
Kernel	PSNR	CN	PSNR	CN
B-spline	8.86	22.16	6.72	97.92
Delta	7.70	1375.75	7.97	53.34
Gaussian	8.84	20.22	6.81	98.90
Rectangular	8.24	423.59	7.20	148.33

- The temporal operator stays the same; represented data and adaptive penalties change.
- Better conditioning is not the same as smaller reconstruction bias.
- Reference-image scores do not identify an absolute intensity baseline from events.

Dissertation: Ch. 2, tab:bump_avg_indoor, tab:bump_avg_outdoor.

The weak equation supplies the natural boundary condition

For every $\boldsymbol{\eta} \in H^1(\Omega;\mathbb{R}^2)$,

$$\int_\Omega (\mathbf{B}\mathbf{v}^* - b)\cdot\boldsymbol{\eta} + \alpha^2 \int_\Omega \nabla\mathbf{v}^* : \nabla\boldsymbol{\eta} = 0.$$

Interior tests give the distributional equation

$$-\alpha^2\Delta\mathbf{v}^* + \mathbf{B}\mathbf{v}^* = b.$$

Since the coefficients imply $\Delta\mathbf{v}^* \in L^2$, Green's identity gives

$$\partial_n\mathbf{v}^* = 0 \quad \text{in } H^{-1/2}(\partial\Omega;\mathbb{R}^2).$$

Takeaway. The theorem uses H^1 fields and weak traces, without claiming a classical solution on a Lipschitz domain.

Why block-Jacobi converges for this graph model

Absorb q_i into $\widehat{\mathbf{B}}_i = q_i \mathbf{B}_i \succeq 0$. Set

$$D_i = \alpha^2 d_i \mathbf{I} + \widehat{\mathbf{B}}_i, \qquad R_{ij} = \alpha^2 w_{ij} \mathbf{I}.$$

For symmetric nonnegative edge weights,

$$\mathbf{z}^\top (D \mp R)\mathbf{z} = \sum_i \mathbf{z}_i^\top \widehat{\mathbf{B}}_i \mathbf{z}_i + \alpha^2 \sum_{\{i,j\}} w_{ij} \left\| \mathbf{z}_i \mp \mathbf{z}_j \right\|^2 .$$

Connectedness and $\sum_i \widehat{\mathbf{B}}_i \succ 0$ eliminate both constant and alternating constant zero modes (the latter on bipartite graphs).

$$D - R \succ 0, \quad D + R \succ 0 \quad \Rightarrow \quad -\mathbf{I} \prec D^{-1/2} R D^{-1/2} \prec \mathbf{I}.$$

Takeaway. The Jacobi iteration matrix is similar to a symmetric matrix with spectral radius strictly below one.

Dissertation: Ch. 3, extension of thm:convergence and cor:convergence_psd_blocks by absorbing q_i into $\widehat{\mathbf{B}}_i$. $\alpha > 0$, $d_i > 0$.

Anisotropy needs a uniform ellipticity floor

For the normalized Nagel–Enkelmann tensor, write $g = \nabla \rho_{\mathrm{evt}}$ and $\gamma_{\mathrm{NE}} > 0$:

$$\mathbf{D} = \frac{g^{\perp}(g^{\perp})^{\top} + \gamma_{\mathrm{NE}}^2 \mathbf{I}}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

Its tangent and normal eigenvalues are

$$1, \qquad \frac{\gamma_{\mathrm{NE}}^2}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

If $g \in L^{\infty}$,

$$\theta \mathbf{I} \preceq \mathbf{D} \preceq \mathbf{I}, \qquad \theta = \frac{\gamma_{\mathrm{NE}}^2}{\|g\|_{L^{\infty}}^2 + \gamma_{\mathrm{NE}}^2} > 0.$$

Takeaway. Diffusion across event-density edges is reduced, while coercivity survives with $\alpha^2\theta$.

Dissertation: Ch. 3, normalized variant of eq:ne_tensor, lem:ne_elliptic, cor:structure_wellposedness.

A variable diffusion tensor must stay inside the divergence

For differentiable $\mathbf{D}$,

$$\nabla\cdot(\mathbf{D}\nabla u) = D_{11}u_{xx} + 2D_{12}u_{xy} + D_{22}u_{yy} \\ + (\partial_x D_{11} + \partial_y D_{12})u_x + (\partial_x D_{12} + \partial_y D_{22})u_y.$$

Take $\rho_{\mathrm{evt}} = x^2/2$, $\gamma_{\mathrm{NE}} = 1$, and $u = x$:

$$\mathbf{D} = \operatorname{diag}\left(\frac{1}{1+x^2}, 1\right), \qquad \nabla\cdot(\mathbf{D}\nabla u) = \frac{-2x}{(1+x^2)^2} \not\equiv 0.$$

All second derivatives of u vanish, so the principal part alone gives the wrong answer.

Dissertation: Ch. 3, rem:aniso_discretization.

The regression radius formula is a heuristic

For full disks on a square lattice of spacing h, with r/h large,

$$Q \sim \frac{\pi r^4}{4h^2}\mathbf{I}, \qquad U_h(r) = \frac{4}{81}L_H^2 r^4 + \frac{8}{\pi}\sigma^2 h^2 r^{-4}.$$

The minimum of this leading expression, when all parameters are positive, is

$$r^* = (162/\pi)^{1/8}(\sigma h/L_H)^{1/4}.$$

- The full disk must lie in a region with the asserted smoothness bound.
- U_h is an asymptotic expression for the finite-design bound, not itself a proved bound on every discrete MSE.
- Affine time surfaces have zero curvature bias; this model supplies no finite balancing radius there.

Dissertation: Ch. 3, prop:ts_regression_bound and Radius selection as a heuristic.

Uniform projection and OLS slope need not be transport partners

The continuous OLS temporal slope over $[s, s+T]$ is

$$\widehat{\beta} = \int_s^{s+T} w_s(t) V_t(t)\, dt, \qquad w_s(t) = \frac{6(t-s)(s+T-t)}{T^3}.$$

Its transport-consistent spatial partner is the same w_s-weighted projection. Pairing it with a uniform projection gives residual

$$\mathbf{v} \cdot \nabla \int_s^{s+T} (T^{-1} - w_s(t)) V(\mathbf{x}, t)\, dt.$$

For $V(x,t) = (2 + x - t)^3$, $\mathbf{v} = (1, 0)$, $s = 0$, $T = 1$, this residual is $1/10$.

Takeaway. Well-posedness of a fixed-coefficient energy does not certify derivative consistency.

Dissertation: Ch. 3, subsec:vox_projection, consistent derivative-operator discussion.

Mean-zero conditioning removes the unobserved mode

In the unweighted scalar model, restrict to

$$\mathcal{V}_0 = \{(\mathbf{f}_L, \mathbf{f}_R) : \mathbf{1}^\top \mathbf{f}_L = \mathbf{1}^\top \mathbf{f}_R = 0\}.$$

The sum–difference transform preserves this space. Its remaining eigenvalues are

$$g_k + \lambda^2, \qquad g_k + \lambda^2 + 2\gamma, \qquad k = 1, \ldots, n-1.$$

Thus

$$\kappa_2(\mathbf{M}|_{\mathcal{V}_0}) = \frac{g_{\max} + \lambda^2 + 2\gamma}{g_1 + \lambda^2} \le \frac{g_{\max} + 2\gamma}{g_1}, \qquad g_1 = 4\sin^2 \frac{\pi}{2n}.$$

Takeaway. The restriction is SPD even at $\lambda = 0$. The uniform-in-λ bound depends on the temporal grid; full-space perturbations remain different.

Dissertation: Ch. 4, cor:mean_zero_conditioning; prop:weighted_conditioning (notes).

The selection theorem has an exact ellipsoid interpretation

Let

$$\mathbf{S} = \tfrac{1}{2}(\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top), \quad \mathbf{T} = \tfrac{1}{4}\mathbf{A}[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}]\mathbf{A}^\top, \quad \mathbf{P} = \mathbf{S} + \mathbf{T} \succ 0.$$

Assume the ideal match $\mathbf{y}_{d_*} = \mathbf{y}_L$. For $\mathbf{e}_d = \mathbf{y}_L - \mathbf{y}_d$,

$$C(d) - C(d_*) = \|\mathbf{e}_d\|_{\mathbf{P}}^2 - 2\mathbf{y}_L^\top \mathbf{S}\mathbf{e}_d = \|\mathbf{y}_d - \mathbf{c}\|_{\mathbf{P}}^2 - \rho^2,$$

$$\mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L, \qquad \rho^2 = \mathbf{y}_L^\top \mathbf{S}\mathbf{P}^{-1}\mathbf{S}\mathbf{y}_L.$$

Takeaway. The ideal match wins if every competitor is outside the ellipsoid. In particular, $\|\mathbf{e}_d\|_{\mathbf{P}} > 2\rho$ is sufficient.

Dissertation: Ch. 4, prop:correct_selection(a), (b2). Candidate-local unweighted model.

An observation perturbation bound can protect a disparity winner

Keep $\mathbf{y}_L$, the candidate set, $\mathbf{A}$, $\mathbf{\Lambda}$, and γ fixed. Perturb each right observation by $\boldsymbol{\xi}_d$. The quadratic candidate cost has

$$\nabla_{\mathbf{y}_d} C = 2\mathbf{P}(\mathbf{y}_d - \mathbf{c}), \qquad \nabla^2_{\mathbf{y}_d} C = 2\mathbf{P}, \qquad \|\mathbf{P}\| \leq \frac{1}{2}.$$

Consequently a uniform bound is

$$\eta = \max_d \left(\|\mathbf{y}_d - \mathbf{c}\| \, \|\boldsymbol{\xi}_d\| + \frac{1}{2} \|\boldsymbol{\xi}_d\|^2 \right).$$

If $\eta < \Delta/2$, the unique winning disparity is preserved.

Takeaway. The ideal correct match may be perturbed too. Changing the left data or adaptive parameters needs additional terms.

Dissertation: Ch. 4, cor:selection_robust.

How the reciprocal IRLS contraction bound is obtained

Write $\mathbf{B}_{\text{time}} = \operatorname{diag}(\mathbf{A}, \mathbf{A})$, $\mathbf{f} = \Phi(\mathbf{g})$, $\mathbf{f}' = \Phi(\mathbf{g}')$, and $\mathbf{r}' = \mathbf{y} - \mathbf{B}_{\text{time}}\mathbf{f}'$. Set $\mathbf{s} = \mathbf{B}_{\text{time}}\mathbf{g}$ and $\mathbf{s}' = \mathbf{B}_{\text{time}}\mathbf{g}'$. Subtract the frozen systems:

$$\mathbf{f} - \mathbf{f}' = \mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\left[\mathbf{W}^{-1/2}(\mathbf{W} - \mathbf{W}')\mathbf{r}'\right].$$

Use

$$\left\|\mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\right\| \le \ell^{-1/2}, \qquad \left\|\mathbf{W}'^{1/2}\mathbf{r}'\right\| \le \left\|\mathbf{W}'^{1/2}\mathbf{y}\right\|.$$

For reciprocal weights, $|w_i - w_i'| \le |s_i - s_i'|w_i w_i'$ and $w_i, w_i' \le \varepsilon^{-1}$.

Takeaway. Combining operator, residual, and weight-increment bounds gives $q = \|\mathbf{A}\|_2 \|\mathbf{y}\| / (\sqrt{\ell}\,\varepsilon^{3/2})$.

Dissertation: Ch. 4, proof of prop:irls_contraction; unclipped reciprocal map.

A converged frozen-weight iteration need not be maximum likelihood

Consider the scalar Gaussian proxy with $y > 0$ and $\varepsilon > 0$, **without a regularizer**:

$$v(\theta) = |\theta| + \varepsilon, \qquad Q_k(\theta) = \frac{(y-\theta)^2}{2v(\theta_k)}.$$

Every frozen quadratic is minimized at $\theta = y$: the IRLS map has that fixed point. The state-dependent negative log-likelihood is

$$\mathcal{L}(\theta) = \frac{(y-\theta)^2}{2v(\theta)} + \frac{1}{2}\log v(\theta).$$

Yet $\mathcal{L}'(y) = 1/[2(y+\varepsilon)] > 0$.

Takeaway. The frozen-weight fixed point is not even stationary for this proxy likelihood.

Dissertation: Ch. 4, prop:irls_vs_nll. Scalar logical illustration outside the positive-$\mathbf{\Lambda}$ standing assumptions.

What the implemented IRLS procedure returns

At every candidate disparity:

1. Initialize from the unweighted exact solve.
2. Recompute predicted-rate weights, **clip for the solve**, and freeze them.
3. Solve the SPD system and damp toward its minimizer.
4. Recompute **unclipped scoring weights**; apply acceptance and stopping rules.
5. Return the last accepted iterate and its score.

The score omits the log-variance term. Masking, averaging, margin normalization, and SGM then change candidate comparisons.

Takeaway. Brouwer and the contraction theorem describe fixed maps; early stopping and aggregation require their own analysis.

Dissertation: Ch. 4, sec:solution, sec:irls_scope; Appendix, app:protocol_ch4.

Depth evaluation distinguishes returned coverage and scored coverage

1. Ground truth is finite and in the reporting depth band.
2. The returned estimate is finite.
3. The estimate also lies in the reporting band.

Configuration	Returned coverage	In-band retention	Scored points
Control	50.17%	51.54%	1.43 M
WLS	49.48%	96.40%	2.66 M

The scoring band is $[0.5, 8]$ m. Out-of-band finite depths are excluded by convention; they are not automatically physically invalid.

Dissertation: Appendix, tab:metrics, sec:success_wls. Complete configurations; method-specific populations.

Pooled kurtosis does not establish heavy-tailed active errors

Suppose a residual distribution mixes a point mass at zero with a zero-mean Gaussian of variance $\sigma^2 > 0$, with zero-mass probability $0 \leq \pi_0 < 1$.

$$\mathbb{E}X^2 = (1-\pi_0)\sigma^2, \qquad \mathbb{E}X^4 = 3(1-\pi_0)\sigma^4.$$

Its population excess kurtosis is

$$\frac{\mathbb{E}X^4}{(\mathbb{E}X^2)^2} - 3 = \frac{3}{1-\pi_0} - 3.$$

- Event-free pixels can have exactly zero fitted transport residual.
- Regularization, unequal variances, and temporal dependence affect diagnostics.
- Test event-active and variance-standardized fitted residuals before choosing a tail model.

Dissertation: Ch. 3, subsec:voxel_failure_analysis; Ch. 5, Model Mismatch in the Residuals.

Adaptive temporal weights are fixed in the theorem

For event activity $a_i = |dc_i|$, Chapter 2 uses these positive rules:

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}} \qquad \text{(sigmoid)}$$

$$\lambda_i = \max\left(10^{-3}, \min\left(\frac{100}{10^{-3} + a_i}, 100\right)\right) \qquad \text{(inverse proportional)}$$

$$\lambda_i = 10^{-3} + 10e^{-\min(a_i, 10)} \qquad \text{(exponential decay)}$$

Chapter 4 instead uses $\lambda_{xy} = \max\{\lambda_{\text{floor}}, \lambda_0 \|\mathbf{b}\| / (\|\mathbf{b}\| + 2)\}$.

Takeaway. Each fixed $\mathbf{\Lambda}$ is covered; the data-dependent map $\mathbf{y} \mapsto \mathbf{\Lambda}(\mathbf{y}) \mapsto \mathbf{f}^*$ is not.

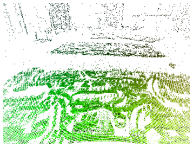
Dissertation: Ch. 2, subsubsec:lambdacal; Ch. 4, sec:solution (regularization terms); Ch. 2, thm:uniqueness.

A real MVSEC flow example is qualitative

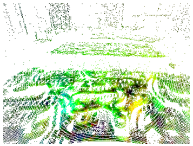
MVSEC indoor_flying2, frame 0340.



Reference



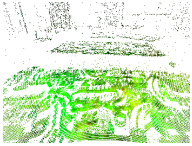
Ground truth



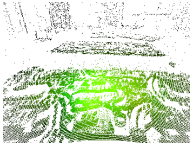
Event-accumulation baseline



Time surface



Voxel



Structure tensor voxel

Takeaway. This is one qualitative example; Chapter 3 reports no aggregate real-data AEE.

Dissertation: Ch. 3, fig:vox_comparison.

Quadratic H^1 regularization and TV answer different questions

The flow energy uses a quadratic gradient penalty:

$$E(\mathbf{v}) = \tfrac{1}{2}\int_\Omega \left(\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}\right) + \tfrac{\alpha^2}{2}\,\|\nabla \mathbf{v}\|_{L^2}^2\,, \qquad \mathbf{v} \in H^1(\Omega;\mathbb{R}^2).$$

- Lax–Milgram gives existence, uniqueness, and coefficient stability under the stated hypotheses.
- TV can preserve jumps, but needs a separate BV existence argument and its own uniqueness and stability assumptions.
- Normalized NE diffusion gives directional quadratic smoothing with a uniform ellipticity floor.

Takeaway. The integrated full-rank condition supports the quadratic H^1 proof; it does not establish a TV theorem.

Dissertation: Ch. 3, thm:unified_wellposedness, function-space remark, eq:ne_tensor.

Frozen prediction weights are not generally an MM majorizer

For a scalar prediction s, let $v(s) = |s| + \varepsilon$, $\varepsilon > 0$, and

$$S(s) = \frac{(y-s)^2}{2v(s)} + R(s), \qquad Q_k(s) = \frac{(y-s)^2}{2v(s_k)} + R(s).$$

The frozen surrogate matches S at s_k, but away from $s = 0$ its derivative omits

$$-\frac{v'(s)(y-s)^2}{2v(s)^2}.$$

Takeaway. Without a tangent derivative, the frozen quadratic is not generally a differentiable tangent majorizer.

Dissertation: Ch. 4, sec:irls_scope, prop:irls_vs_nll; derivative consequence, not a stated MM theorem.

A Gaussian model gives an evidence interpretation of attenuation

For common fixed $\mathbf{\Lambda} \succ 0$, $\gamma \geq 0$, and independent unit Gaussian noise, the candidate conditional is

$$\mathbf{y}_R \mid \mathbf{y}_L \sim \mathcal{N}(\mathbf{c}, (2\mathbf{P})^{-1}), \qquad \mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L.$$

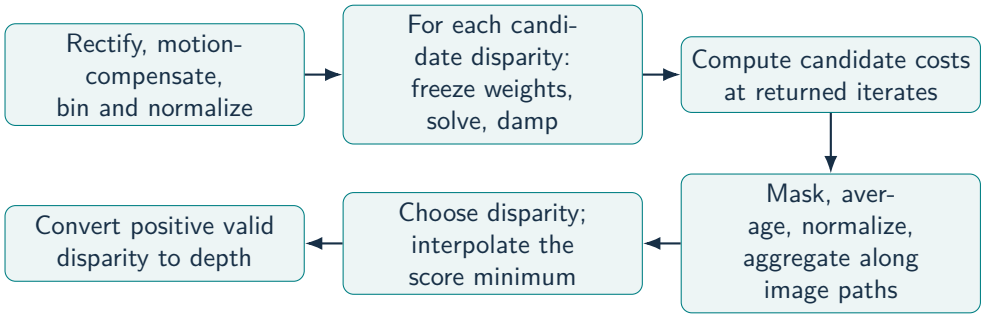
Coupled versus independent evidence satisfies

$$-\log \frac{p_\gamma(\mathbf{y}_L, \mathbf{y}_R)}{p_0(\mathbf{y}_L, \mathbf{y}_R)} = D(\mathbf{y}_L - \mathbf{y}_R) + \text{constant}.$$

Takeaway. The conditional mean explains attenuation; the discrepancy alone is an unevaluated alternative score.

Dissertation: Supplementary Gaussian derivation from Ch. 4, lem:matching_cost_closed_form, prop:correct_selection.

The implemented estimator has several distinct layers



Theorems describe specific layers; empirical evaluation describes the complete estimator.

Takeaway. The exact minimum of a computed score list is not a certificate of the true disparity or of a global image-level energy minimum.

Dissertation: Ch. 4, sec:solution; Appendix, app:protocol_ch4.

What does the dissertation contribute mathematically?

- A transparent temporal inversion with an explicit observation null space and quantitative fixed-weight guarantees.
- One well-posedness and stability theorem for several event-derived quadratic flow energies, with representation-specific hypotheses.
- A distinction between a coercive spline regularizer and a nonconvex contrast objective, plus full-objective existence.
- Stereo block structure, profiled-cost bias, ideal selection and robustness criteria, and a scoped adaptive-weight analysis.

Takeaway. Classical proof tools connect the models; the event-specific formulation, hypotheses, consequences, and limitations are the research content.

Dissertation: Chs. 1 and 5; labeled results throughout Chs. 2–4.

Literature anchors and where to find the claims

- **Sensor and field overview:** Gallego et al., *Event-based Vision: A Survey*; dissertation bibliography key `gallego2022event`.
- **Variational flow:** Horn and Schunck, *Determining Optical Flow*; `horn1981determining`.
- **Timestamp geometry:** Benosman et al., *Event-Based Visual Flow*; `benosman2014event`.
- **Contrast maximization:** Gallego, Rebecq, and Scaramuzza, *A Unifying Contrast Maximization Framework*; `gallego2018unifying`.
- **Evaluation data:** Zhu et al., *The Multivehicle Stereo Event Camera Dataset*; `MVSEC2018`.

Every slide's footer identifies its dissertation source. The companion source map records theorem labels, current theorem numbers, figures, numerical tables, and scope.

Dissertation: GMU-Diss_template.tex; gmuETD.bib; Ch. 3 related work; Appendix evaluation protocol.

Stable estimates can still have regularization bias

For fixed scalar $\lambda > 0$, $\mathbf{y} = \mathbf{A}\mathbf{f}^\dagger + \boldsymbol{\eta}$, and $\mathbf{K} = \mathbf{A}^\top\mathbf{A} + \lambda^2\mathbf{I}$,

$$\mathbf{f}_\lambda - \mathbf{f}^\dagger = -\lambda^2\mathbf{K}^{-1}\mathbf{f}^\dagger + \mathbf{K}^{-1}\mathbf{A}^\top\boldsymbol{\eta}.$$

The two terms are regularization bias and propagated observation noise.

Takeaway. A stable estimate need not equal the physical signal.

Dissertation: Ch. 2, thm:uniqueness, prop:stability; direct bias-noise identity.

The physical event stream loses more than one baseline

The binned increment matrix has $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$, but the ideal sensor can lose time-varying signals too.

$$L_1(t) = L_0, \qquad L_2(t) = L_0 + a\sin(2\pi t/T), \qquad 0 < a < c_{\mathrm{th}}, \quad t \in [0, T].$$

Both start at L_0, emit no events, and have positive intensities $I_j(t) = e^{L_j(t)}$.

Takeaway. The chain nullspace describes the discrete increment surrogate, not every ambiguity of the physical sensor.

Dissertation: Ch. 2, eq:polarity, lem:nullspace; direct ideal-sensor example.

Integrated directional information is necessary for uniqueness

Let Ω be connected, bounded, and Lipschitz; $\alpha > 0$; $\mathbf{B} \in L^\infty$ symmetric PSD; and $b \in L^2$. Minimize over unrestricted $H^1(\Omega; \mathbb{R}^2)$ fields. Set $M_B = \int_\Omega \mathbf{B}$. If M_B is singular, choose $c \neq 0$ with $M_B c = 0$. Then $\mathbf{B}c = 0$ a.e. and

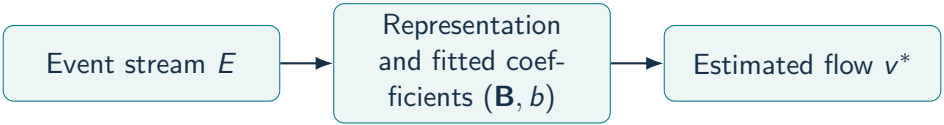
$$E(\mathbf{v} + tc) = E(\mathbf{v}) - t\, c^\top \int_\Omega b.$$

If the coefficient is nonzero, E is unbounded below; if zero, it is flat in direction c.

Takeaway. The integrated matrix condition is necessary for a unique minimizer in this unrestricted quadratic model.

Dissertation: Ch. 3, thm:unified_wellposedness; direct converse for its quadratic energy.

Coefficient stability is one stage of event-to-flow stability



The theorem controls coefficient-to-flow sensitivity. A Lipschitz representation map could compose with that bound.

Takeaway. An end-to-end guarantee also needs a metric and stability result for event preprocessing.

Dissertation: Ch. 3, thm:unified_wellposedness, prop:ts_regression_bound; supplementary scope explanation.

Refinement requires a choice of continuum scaling

Illustrative smooth, uniform-time model only: spacing h, increments $y_i = hg_i$, and target energy

$$\frac{1}{2}\int |f' - g|^2 dt + \frac{\beta_{\text{time}}^2}{2}\int |f|^2 dt \quad \rightsquigarrow \quad \frac{1}{2h}\|\mathbf{A}f - y\|^2 + \frac{\beta_{\text{time}}^2 h}{2}\|f\|^2 .$$

Multiplying the discrete objective by h gives the Chapter 2 form with

$$\lambda_h = \beta_{\text{time}} h.$$

Takeaway. The penalty's grid scaling depends on the continuum objective being approximated.

Dissertation: Ch. 2, eq:tempdiffA, eq:in; supplementary smooth-model scaling illustration.

Strong smoothness selects one constant flow

For fixed $\Omega, \mathbf{B}, b$ satisfying the unified theorem, with unrestricted H^1 fields and the same α^2 gradient coefficient,

$$\alpha \to \infty \quad \Longrightarrow \quad v_\alpha \to c \text{ in } H^1, \qquad c = \left(\int_\Omega \mathbf{B} \right)^{-1} \int_\Omega b.$$

Takeaway. Very strong spatial regularization removes variation and leaves the data-balanced constant field.

Dissertation: Ch. 3, thm:unified_wellposedness; direct strong-regularization limit.

Transporting a scalar differs from conserving a density

Scalar transport and source-free density conservation are

$$V_t + v \cdot \nabla V = 0, \qquad V_t + \nabla \cdot (Vv) = 0, \qquad \nabla \cdot (Vv) = v \cdot \nabla V + V \nabla \cdot v.$$

They coincide when $V \nabla \cdot v = 0$, in particular for divergence-free flow.

Takeaway. The continuity equation adds a density-compression term to scalar transport.

Dissertation: Ch. 3, subsec:vox_projection, sec:vox_cont_exclusion; direct product-rule comparison.

Correlation changes uncertainty without removing coercivity

For a centered symmetric regression design Z with $Q = Z^\top Z \succ 0$, timestamp-error covariance Σ gives

$$\mathrm{Cov}(\hat{g}) = Q^{-1} Z^\top \Sigma Z Q^{-1}.$$

This reduces to $\sigma^2 Q^{-1}$ when $\Sigma = \sigma^2 I$.

With any $\Gamma \succ 0$, the temporal system matrix is $\mathbf{A}^\top \Gamma^{-1} \mathbf{A} + \mathbf{\Lambda}^2 \succ 0$ when $\mathbf{\Lambda} \succ 0$.

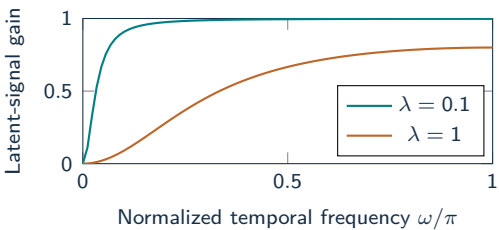
Takeaway. The deterministic quadratic solve stays coercive, while statistical uncertainty depends on the error covariance.

Dissertation: Ch. 3, prop:ts_regression_bound; Ch. 2, eq:in, thm:uniqueness; direct correlated-error extension.

What the penalty does to temporal information

For scalar $\Lambda = \lambda \mathbf{I}$ and noiseless data $\mathbf{y} = \mathbf{A}\mathbf{f}_{\mathrm{true}}$:

$$g_k = 4 \sin^2 \frac{\pi k}{2n}, \qquad H_k = \frac{g_k}{g_k + \lambda^2}, \qquad H_0 = 0.$$



The baseline is removed.

Low temporal modes are attenuated most.

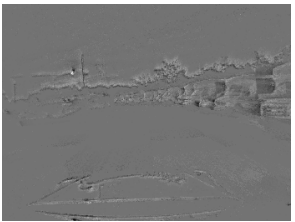
A unique estimate does not restore the missing physical baseline.

Dissertation: Ch. 4, lem:A_spectrum, thm:frequency_response. Curves: continuous extension.

Temporal evidence: structure without an absolute baseline



Reference



Spline reconstruction

MVSEC indoor kernel comparison

Kernel	PSNR	CN
B-spline	8.86	22.16
Delta	7.70	1375.75
Gaussian	8.84	20.22

Takeaway. Kernel and adaptive-weight choices affect both the reconstruction and the solved system.

Dissertation: Ch. 2, tab:bump_avg_indoor and outdoor bump-sweep gallery. CN: reported condition number.